

Lyapunov-Controlled Minimax Policy Learning for Adversarial Dynamic Spectrum Access

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Abstract—Adaptive jamming makes dynamic spectrum access (DSA) adversarial because the transmitter and jammer adapt their channel-selection policies. We model it as a zero-sum Markov game and propose Lyapunov-Controlled Minimax Policy Learning (LCMPL) with separate neural policies. LCMPL combines the simultaneous minimax policy-gradient field with a Jacobian–field correction, decouples their step sizes, and jointly regulates them using local game geometry and a communication-aware merit. A curvature identity identifies when the correction reduces the saddle-field energy that measures joint policy stationarity. Under local smoothness, geometric coverage by the field and curvature directions, and bounded learning and local-model errors, the time-averaged expected stationarity residual has an $O(1/K)$ transient plus a residual term, where K is the number of joint transmitter–jammer updates. Under a local policy-gap condition, a communication corollary converts this learning-stability result into a bound on the transmitter’s worst-case spectral-efficiency loss. Across the nominal and trajectory-sampled benchmarks, LCMPL attains the highest final mean worst-case spectral-efficiency return among the compared methods. Trajectory-sampled experiments show a consistent rate gain over the field-only ablation, with a positive paired 95% confidence interval.

Index Terms—Anti-jamming communications, dynamic spectrum access, reinforcement learning, minimax policy learning, zero-sum Markov games.

I. INTRODUCTION

Dynamic spectrum access (DSA) lets a secondary transmitter select a channel based on incumbent activity and channel conditions [1], while accounting for fading, interference, switching overhead, and adaptive jamming [2], [3], [4].

Each learned policy changes the opponent’s future observations and policy gradient. Adversarial DSA therefore combines exogenous radio variation, endogenous opponent adaptation, and sampled learning uncertainty. The central problem is to stabilize joint policy adaptation without sacrificing spectral efficiency.

A two-player zero-sum Markov game captures this interaction [5], [6]. The transmitter policy π_θ , with parameter vector θ , maximizes a discounted spectral-efficiency return, while the jammer policy ν_ψ , with parameter vector ψ , minimizes it through $\max_\theta \min_\psi J(\theta, \psi)$. Their policy gradients form a saddle field; worst-case transmitter return measures communication robustness, while exploitability measures sensitivity to further policy adaptation.

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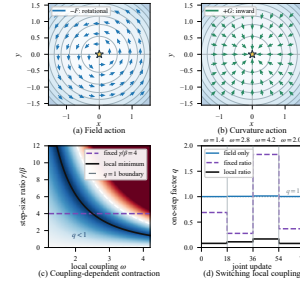


Fig. 1. Why saddle-field step sizes must adapt to local coupling. For the local update $z_+ = z - \beta F_\omega(z) + \gamma \nabla F_\omega(z) F_\omega(z)$, F_ω is the saddle field at coupling strength ω , β and γ are the field and curvature step sizes, and $q = \|z_+\|/\|z\|$ is the one-step norm ratio. Panels (a) and (b) show the rotational field action and inward Jacobian–field action. Panel (c) shows that the contraction region $q < 1$ and its locally minimizing curvature-to-field ratio γ/β vary with ω . Panel (d) follows switching coupling: a fixed ratio crosses $q = 1$, whereas the local ratio remains contractive.

Fig. 1 isolates the resulting learning mechanism. In a bilinear local model, simultaneous gradient descent–ascent (GDA) follows a rotational field action and can cycle, while the Jacobian–field action supplies radial contraction [7]. The contraction region depends on both local coupling and the field-to-curvature step-size ratio; when the wireless state or either neural policy changes that coupling, a fixed ratio can move from contraction to expansion. The proximal point method (PPM) evaluates the field implicitly, whereas the extragradient method (EGM) uses an extrapolated pair [8], [9]. Their local expansions include a Jacobian–field action but couple the effective field and curvature scales through one learning rate [10]. Adaptive Mirror-Prox methods use observed operator geometry to tune extragradient updates across smoothness regimes [11], while double-step extragradient assigns distinct scales to the extrapolation and update stages [12]. Our formulation instead treats the policy-gradient field and its Jacobian–field correction as separate search coordinates whose local coupling changes with the wireless state and neural policies.

Field-energy reduction alone cannot distinguish a robust high-rate policy from a low-rate stationary policy. Guided by adaptive learning-rate and Lyapunov-drift control [13], [14], we coordinate geometric stabilization with worst-case-rate loss and exploitability.

We develop Lyapunov-Controlled Minimax Policy Learning (LCMPL) for adaptive anti-jamming DSA. Its two-variable

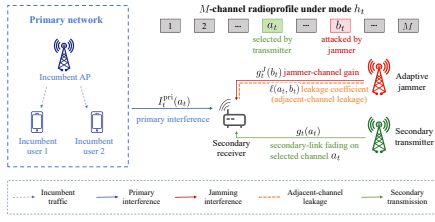


Fig. 2. Adversarial wireless environment for anti-jamming dynamic spectrum access.

quadratic program chooses separate field and curvature step sizes from a local communication-aware Lyapunov model, while a same-state test filters detrimental curvature. We prove an $O(1/K)$ stationarity transient with a learning-error residual and, under a local policy-gap condition, a worst-case-rate bound.

II. ADVERSARIAL DSA SYSTEM AND MARKOV-GAME MODEL

A. Multi-Channel Wireless Environment and Stage Utility

Fig. 2 shows an incumbent network, a secondary link, and an adaptive jammer over M licensed subchannels. The discrete variable $h_t \in \mathcal{H} = \{1, \dots, L\}$ labels the current channel-occupancy or radio mode. It is neither a selected channel nor a scalar gain; it indexes the complete channel-condition profile that applies at time t . Conditional on this mode, $g_t(a)$ is the secondary-link gain on channel a , while I_t collects the incumbent-interference profile $I_t^{\text{pri}}(a)$ and jammer-receiver gains $g_t^J(b)$. The transmitter and jammer simultaneously choose channels $a_t \in \mathcal{A}$ and $b_t \in \mathcal{B}$, where $\mathcal{A} = \mathcal{B} = \{0, \dots, M-1\}$. The observed Markov state is $s_t = (h_t, g_t, I_t, a_{t-1}) \in \mathcal{S}$; the previous transmitter channel a_{t-1} is included because changing channels incurs a switching cost. In the reported finite benchmark, the background interference and jammer-receiver gain are fixed, and each mode h_t selects a distinct secondary-link gain vector $g(h_t)$. The pair consisting of $g(h_t)$ and one-hot a_{t-1} therefore gives both neural policies a one-to-one encoding of the LM compact states.

The constants P_s , P_J , and N_0 denote secondary transmission power, jammer interference level, and receiver noise power, respectively. The factor $\ell(a, b)$ models co-channel, adjacent-channel, and nonadjacent leakage, and $c_{\text{sw}} \geq 0$ is the switching cost. For actions (a_t, b_t) , the received signal-to-interference-plus-noise ratio (SINR) and switching-cost-adjusted stage spectral efficiency are

$$\text{SINR}_t = \frac{P_s g_t(a_t)}{N_0 + I_t^{\text{pri}}(a_t) + P_J g_t^J(b_t) \ell(a_t, b_t)}, \quad (1)$$

$$r_t = \log_2(1 + \text{SINR}_t) - c_{\text{sw}} \mathbf{1}_{\{a_t \neq a_{t-1}\}}.$$

Here, $\mathbf{1}_{\{\cdot\}}$ is the indicator function.

B. Radio Dynamics and Minimax Policy Objective

The radio mode follows an action-independent correlated Markov chain with transition matrix P_h . Writing (h', a') for the next compact state, the benchmark transition is

$$P((h', a') | (h, a_{\text{prev}}, a, b) = P_h(h'|h) \mathbf{1}_{\{a'=a\}}.$$

We use $P_h(h|h) = \rho_h$ and $P_h(h'|h) = (1 - \rho_h)/(L - 1)$ for $h' \neq h$, where ρ_h controls temporal correlation. The reported benchmarks set $L = M$ and initialize uniformly over the LM compact states. This specialization makes the exogenous chain control correlated secondary-link fading, while the adaptive jammer creates endogenous interference through its selected channel and leakage profile. The wireless model above specifies the physical interaction. We next define the learning objective and distinguish it from the communication quantity used for evaluation. Let $\pi_{\theta}(a|s)$ and $\nu_{\psi}(b|s)$ denote differentiable transmitter and jammer policies. The physical normalized discounted spectral-efficiency return is

$$J(\theta, \psi) = (1 - \delta) \mathbb{E} \left[\sum_{t=0}^{\infty} \delta^t r_t \right],$$

where $\delta \in (0, 1)$ is the discount factor. To discourage either softmax policy from concentrating before it has adapted to a changing radio mode or opponent, we train with the entropy-regularized return

$$J_{\tau} = J + (1 - \delta) \tau \mathbb{E} \left[\sum_{t \geq 0} \delta^t \{H(\pi_{\theta}(\cdot|s_t)) - H(\nu_{\psi}(\cdot|s_t))\} \right],$$

where $\tau > 0$ is the entropy weight and $H(\cdot)$ is the Shannon entropy. The entropy terms encourage channel exploration and smooth the policy gradients and best-response quantities used by the controller. All training directions and the Lyapunov model use J_{τ} ; the rate acceptance test and reported communication metrics use the physical return J in bit/s/Hz. This division gives the learner smooth exploratory gradients without redefining the communication objective. To test robustness beyond the current neural opponent, let Π_T and Π_J be the full stationary Markov policy spaces. The primary metric is the learned transmitter's worst-case return over Π_J ,

$$R_{\text{BR}}(\theta) = \min_{\nu \in \Pi_J} J(\pi_{\theta}, \nu).$$

The unregularized exploitability diagnostic measures the remaining improvement available to either player:

$$\Delta(z) = \max_{\pi \in \Pi_T} J(\pi, \nu_{\psi}) - \min_{\nu \in \Pi_J} J(\pi_{\theta}, \nu).$$

Replacing J by J_{τ} defines the smooth regularized gap Δ_{τ} used only inside the training merit. Dynamic programming computes R_{BR} , Δ , and its soft counterpart Δ_{τ} . With the communication objective fixed, the next section addresses the instability of simultaneous transmitter-jammer adaptation.

III. GEOMETRY-AWARE MINIMAX POLICY LEARNING

A. Curvature-Induced Two-Direction Learning

For the joint parameter $z = (\theta, \psi)$, define the saddle field

$$\mathbf{F}(z) = \begin{bmatrix} -\nabla_{\theta} J_{\tau}(\theta, \psi) \\ \nabla_{\psi} J_{\tau}(\theta, \psi) \end{bmatrix}. \quad (2)$$

For a learning rate $s > 0$, GDA uses $z_+ = z - s \mathbf{F}(z)$ for transmitter ascent and jammer descent. The field energy $\phi(z) = \|\mathbf{F}(z)\|^2 / 2$ measures joint policy stationarity. Under GDA,

$\phi(\mathbf{z}_+) = \phi(\mathbf{z}) - s\langle \nabla \phi, \mathbf{F} \rangle + O(s^2)$; pure rotation has zero first-order change despite $\mathbf{F} \neq 0$. PPM and EGM respond to this field variation by evaluating the field after a policy displacement:

$$\begin{aligned} \text{GDA: } \mathbf{z}_+^{\text{GDA}} &= \mathbf{z} - s\mathbf{F}(\mathbf{z}), \\ \text{PPM: } \mathbf{z}_+^{\text{PPM}} &= \mathbf{z} - s\mathbf{F}(\mathbf{z}_+^{\text{PPM}}), \\ \text{EGM: } \tilde{\mathbf{z}} &= \mathbf{z} - s\mathbf{F}(\mathbf{z}), \quad \mathbf{z}_+^{\text{EGM}} = \mathbf{z} - s\mathbf{F}(\tilde{\mathbf{z}}), \end{aligned}$$

Here, $\tilde{\mathbf{z}}$ is EGM's extrapolation point; PPM uses an implicit next-policy gradient, while EGM uses an explicit predictor. Let $\nabla \mathbf{F}(\mathbf{z})$ denote the Jacobian of the saddle field. Expanding the displaced field in EGM, or the implicit field in PPM, gives $\mathbf{z}_+ = \mathbf{z} - s\mathbf{F}(\mathbf{z}) + s^2 \nabla \mathbf{F}(\mathbf{z}) \mathbf{F}(\mathbf{z}) + O(s^3)$. Their common correction motivates the Jacobian-field direction

$$\mathbf{G}(\mathbf{z}) = \nabla \mathbf{F}(\mathbf{z}) \mathbf{F}(\mathbf{z}). \quad (3)$$

The term $+s^2 \mathbf{G}$ is the shared local correction that distinguishes PPM and EGM from the rotational field step. It points inward in the bilinear model of Fig. 1 and motivates treating $-\mathbf{F}$ and $+\mathbf{G}$ as separate update coordinates. PPM and EGM couple their effective scales through the fixed relation (s, s^2) , although the useful balance changes with local coupling. LCMPL instead assigns the field and curvature coordinates separate step sizes β_k and γ_k :

$$\mathbf{z}_{k+1} = \mathbf{z}_k - \beta_k \mathbf{F}_k + \gamma_k \mathbf{G}_k. \quad (4)$$

Here, $\mathbf{F}_k$ and $\mathbf{G}_k$ denote the field and Jacobian-field directions available at update k . Population experiments evaluate them exactly, while trajectory experiments estimate both directions from one on-policy batch. The curvature direction introduces no additional wireless action; it modifies only the policy update using local game geometry. The remaining design problem is to choose β_k and γ_k without trading policy stabilization for communication performance.

B. Communication-Aware Lyapunov Policy Update

A small field energy identifies joint policy stationarity but does not distinguish a robust high-rate policy from a low-rate stationary policy. LCMPL therefore evaluates both directions through a Lyapunov merit that combines field energy, exploitability, and worst-case-rate deficit. Define the normalized communication loss

$$\mathcal{L}_c(\mathbf{z}) = \lambda_\Delta \left(\frac{\Delta_\tau(\mathbf{z})}{\Delta_{\tau,0}} \right)^2 + \lambda_R \left(\frac{D_{R,\tau}(\boldsymbol{\theta})}{D_0} \right)^2,$$

where $\lambda_\Delta, \lambda_R \geq 0$ are merit weights and $\Delta_{\tau,0}, D_0 > 0$ are initial scales. The first term penalizes a policy pair that remains exploitable, while the second penalizes loss of robust communication rate. Let $r_{\max}$ be the largest stage reward in the finite game and define the entropy-regularized stage-return ceiling $R_{\text{ub},\tau} = r_{\max} + \tau \log M$. The deficit $D_{R,\tau} = R_{\text{ub},\tau} - \min_{\nu \in \Pi_J} J_\tau(\pi_\theta, \nu) \geq 0$ uses this ceiling for normalization. The Lyapunov merit is

$$V(\mathbf{z}) = \frac{\phi(\mathbf{z})}{\phi_0} + \mathcal{L}_c(\mathbf{z}), \quad (5)$$

where $\phi_0 > 0$ is the initial field-energy scale. The merit controls training but does not replace the physical-rate metric. Its one-step drift provides a common criterion for geometric stabilization and communication quality.

Let $\mathcal{F}_k$ contain the learning history before update k . Direct evaluation over the full neural parameter space is unnecessary because the update lies in the two-dimensional span of $\mathbf{d}_{k,1} = -\mathbf{F}_k$ and $\mathbf{d}_{k,2} = \mathbf{G}_k$. For a candidate $\mathbf{u} = (\beta, \gamma)^\top$, define its conditional Lyapunov drift by

$$\mathcal{D}_k(\mathbf{u}) = \mathbb{E}[V(\mathbf{z}_k + \beta \mathbf{d}_{k,1} + \gamma \mathbf{d}_{k,2}) - V(\mathbf{z}_k) \mid \mathcal{F}_k].$$

A negative $\mathcal{D}_k(\mathbf{u})$ means that the candidate step decreases the expected merit given the information available before update k . Minimizing this drift therefore balances three goals in one decision: reducing the saddle-field energy, limiting exploitability, and preserving the transmitter's worst-case rate. Because $V \geq 0$, such one-step decreases can also be summed over time, which is the basis of the finite-time bound in Section IV. Directly minimizing $\mathcal{D}_k$ would require repeated evaluation of the nonlinear policy objective for every step-size pair. Instead, a directional Taylor expansion gives

$$\begin{aligned} \mathcal{D}_k(\mathbf{u}) &= \sum_{i=1}^2 u_i \mathbb{E}[\langle \nabla V(\mathbf{z}_k), \mathbf{d}_{k,i} \rangle \mid \mathcal{F}_k] \\ &\quad + \frac{1}{2} \sum_{i,j=1}^2 u_i u_j \mathbb{E}[\mathbf{d}_{k,i}^\top \nabla^2 V(\mathbf{z}_k) \mathbf{d}_{k,j} \mid \mathcal{F}_k] + \rho_k(\mathbf{u}). \end{aligned}$$

where ρ_k collects higher-order and direction-estimation errors. LCMPL approximates the two directional slopes and curvatures in this expansion and absorbs the resulting discrepancy, including ρ_k , into the model-error term used in Section IV. This yields the local surrogate

$$\begin{aligned} m_k(\mathbf{u}) &= \mathbf{a}_k^\top \mathbf{u} + \frac{1}{2} \mathbf{u}^\top \mathbf{H}_k \mathbf{u}, \\ \tilde{\mathbf{u}}_k &= \arg \min_{\mathbf{u} \in \mathcal{U}_k} m_k(\mathbf{u}), \quad \mathcal{U}_k = [0, \bar{\beta}] \times [0, \bar{\gamma}_k], \end{aligned} \quad (6)$$

where $\mathbf{a}_k \in \mathbb{R}^2$ contains directional slopes, $\mathbf{H}_k \succeq h_{\min} \mathbf{I}$ is a regularized diagonal directional-curvature estimate, $h_{\min} > 0$, and $\bar{\beta}, \bar{\gamma}_k > 0$ cap the two step sizes. The box keeps the Taylor model local, while the positive curvature regularization makes the two-variable problem well posed. The quadratic program (QP) therefore replaces an expensive search over the exact drift by a tractable prediction of the best local merit decrease. When rotational coupling makes the field coordinate ineffective, a negative curvature slope can activate γ_k and move the update inward. When the field coordinate already decreases the merit, the same model can reduce or suppress the curvature step. LCMPL compares the proposed pair with the field-only minimizer of the same model and accepts curvature only when the realized merit and physical worst-case rate are no worse. LCMPL-F fixes $\gamma_k = 0$ while retaining the same field-direction merit, local model, and acceptance rule. Fig. 3 summarizes how communication feedback determines the two directions, the Lyapunov-drift model selects their step sizes through a box QP, and the same-state rate test protects communication performance.

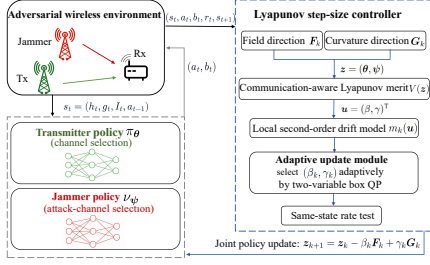


Fig. 3. LCMPL minimax policy-learning framework. Communication feedback determines the field and curvature directions. A local Lyapunov-drift model selects their step sizes, and a same-state rate test protects the communication objective.

IV. LEARNING STABILITY AND COMMUNICATION GUARANTEES

This section connects curvature-induced field-energy decrease to the implemented update and its finite-time communication guarantee. Write $\nabla F = S + W$, where $S = (\nabla F + \nabla F^T)/2$ and $W = (\nabla F - \nabla F^T)/2$.

Proposition 1 (Curvature identity for adversarial policy learning): At every point where F is differentiable, the Jacobian–field direction satisfies

$$\langle \nabla \phi, G \rangle = \|SF\|^2 - \|WF\|^2. \quad (7)$$

Hence, $+G$ decreases the field energy when $\|WF\| > \|SF\|$.

Proof: With $A = \nabla F$, $\langle \nabla \phi, G \rangle = F^T A^2 F$. Expanding $A = S + W$ eliminates the cross term because $SW + WS$ is antisymmetric. Moreover, $F^T S^2 F = \|SF\|^2$ and $F^T W^2 F = -\|WF\|^2$, which gives Eq. (7). $\square$

Proposition 1 explains why the curvature coordinate can reduce field energy. In particular, antisymmetric coupling creates the rotation of the field step and the inward action of the Jacobian–field correction. The next conditions connect that ideal geometry to the local model and accepted learning update. Define $D_F(z) = \langle \nabla V(z), F(z) \rangle$ and $D_G(z) = -\langle \nabla V(z), G(z) \rangle$, the first-order decreases of V along $-F$ and $+G$. Write $\Delta V_k = V(z_{k+1}) - V(z_k)$.

Assumption 1 (Local regularity and model accuracy):

- (a) A compact set $\mathcal{T}$ contains the accepted iterates and candidate segments, and $F \in C^1$ and $V \in C^2$ on a neighborhood of $\mathcal{T}$.
- (b) For a fixed $s > 0$, both pure-coordinate pairs se_1 and se_2 belong to $\mathcal{U}_k$, where e_i is the i th coordinate vector.
- (c) Finite nonnegative $\mathcal{F}_k$ -measurable errors ϵ_k^{acc} and ϵ_k^{mdl} satisfy

$$\begin{aligned} \mathbb{E}[\Delta V_k | \mathcal{F}_k] &\leq \mathbb{E}[m_k(\tilde{u}_k) | \mathcal{F}_k] + \epsilon_k^{\text{acc}}, \\ \mathbb{E}[m_k(se_i) | \mathcal{F}_k] &\leq -sD_i(z_k) + \epsilon_k^{\text{mdl}}, \quad i \in \{1, 2\}, \end{aligned}$$

where $D_1 = D_F$ and $D_2 = D_G$.

The two errors aggregate sampled policy-gradient and Jacobian-action noise, the directional-model approximation, and the acceptance step. In population updates they record local-model and acceptance mismatch; in trajectory updates they also include sampling error.

Define the population field-energy decrease signals

$$\chi_F(z) = F^T S F, \quad \chi_G(z) = -\langle \nabla \phi(z), G(z) \rangle.$$

Positive χ_F and χ_G certify decrease of ϕ along $-F$ and $+G$, respectively.

Assumption 2 (Geometric coverage): The local coverage modulus

$$m_{\text{geo}} = \inf_{z \in \mathcal{T}} \frac{\max\{\chi_F(z), \chi_G(z)\}}{\|F(z)\|^2} \quad (8)$$

satisfies $m_{\text{geo}} > 0$.

Geometric coverage does not require the field coordinate to be a strict descent direction; it requires at least one of the two coordinates to decrease field energy at a rate proportional to $\|F\|^2$. Let $\sigma_{\min}$ denote the smallest singular value. At a target stationary point z^* , the conditions $S(z^*) \succeq 0$ and $\sigma_{\min}(\nabla F(z^*)) > 0$ imply Eq. (8) on a sufficiently small neighborhood by compactness of the unit sphere and continuity; this includes nonsingular pure rotation, where $\chi_F = 0$ and $\chi_G > 0$. Compactness and smoothness make

$$B_c = \sup_{z \in \mathcal{T}} \max\{|\langle \nabla \mathcal{L}_c, F \rangle|, |\langle \nabla \mathcal{L}_c, G \rangle|\} < \infty.$$

Because $V = \phi/\phi_0 + \mathcal{L}_c$, the larger of the two directional signals satisfies

$$\begin{aligned} \max\{D_F(z), D_G(z)\} &\geq \mu_{\text{geo}} \|F(z)\|^2 - B_c, \\ \mu_{\text{geo}} &= m_{\text{geo}}/\phi_0. \end{aligned} \quad (9)$$

Define the total conditional learning residual

$$\mathcal{E}_k(s) = sB_c + \epsilon_k^{\text{acc}} + \epsilon_k^{\text{mdl}}. \quad (10)$$

This single term separates the geometric benefit of the two directions from stochastic sampling, local approximation, and acceptance errors.

Proposition 2 (Conditional Lyapunov drift): Under Assumptions 1–2, the accepted update satisfies

$$\begin{aligned} \mathbb{E}[V(z_{k+1}) - V(z_k) | \mathcal{F}_k] \\ \leq -s\mu_{\text{geo}} \|F(z_k)\|^2 + \mathcal{E}_k(s). \end{aligned}$$

Proof: The QP optimum is no larger than either pure-coordinate comparator in Assumption 1(b). Assumption 1(c) transfers this model dominance to the accepted Lyapunov change, and Eq. (9) supplies the larger population decrease. $\square$

Theorem 1 (Main finite-time minimax stationarity result): Under Assumptions 1–2, every integer $K \geq 1$ satisfies

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|F(z_k)\|^2 \leq \frac{V(z_0)}{s\mu_{\text{geo}}K} + \frac{\bar{\mathcal{E}}_K(s)}{s\mu_{\text{geo}}}, \quad (11)$$

where $\bar{\mathcal{E}}_K = K^{-1} \sum_{k=0}^{K-1} \mathbb{E} \mathcal{E}_k$.

Proof: Take total expectations in Proposition 2, sum over k , and use $V(z_K) \geq 0$. $\square$

If $\sup_k \mathbb{E} \mathcal{E}_k(s) \leq \bar{E} < \infty$, Theorem 1 gives an $O(1/K)$ transient plus the floor $\bar{E}/(s\mu_{\text{geo}})$. It covers pure rotation: the field has zero first-order effect, while curvature supplies

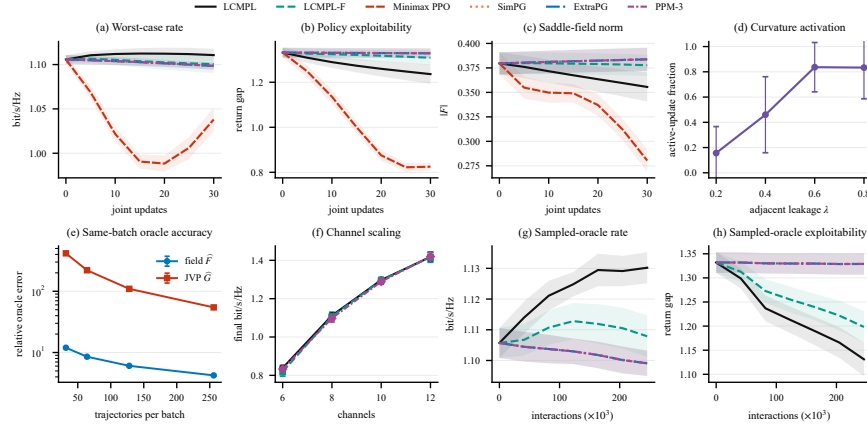


Fig. 4. Adversarial DSA results: (a)–(c) nominal population dynamics; (d) leakage-dependent curvature activation; (e) sampled-direction fidelity; (f) channel scaling; and (g)–(h) equal-interaction trajectory learning. Means use 12 seeds except eight in (f); bands in (a)–(c) and (g)–(h) show one standard error, and bars in (d)–(f) show 95% confidence intervals.

the strict decrease in Fig. 1 and Proposition 1. Here, s is an analytical comparator, not an implemented learning rate.

The following local compatibility condition bridges Theorem 1 from neural parameter-space stationarity to full-policy best responses.

Let $v^* = \max_{\pi \in \Pi_T} \min_{\nu \in \Pi_J} J(\pi, \nu)$ denote the unregularized stationary-policy game value. Since each M -ary policy has entropy at most $\log M$, the regularized and unregularized returns differ by at most $\tau \log M$ for either player. Consequently,

$$0 \leq v^* - R_{\text{BR}}(\theta) \leq \Delta_\tau(z) + 2\tau \log M. \quad (12)$$

Let B_K denote the right-hand side of Eq. (11).

Corollary 1 (Main communication result: worst-case spectral efficiency): If nonnegative constants C_{gap} and ξ_{gap} satisfy $\Delta_\tau(z) \leq C_{\text{gap}} \|\mathbf{F}(z)\|^2 + \xi_{\text{gap}}$ on $\mathcal{T}$, then

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}[v^* - R_{\text{BR}}(\theta_k)] \leq C_{\text{gap}} B_K + \xi_{\text{gap}} + 2\tau \log M.$$

Proof: Average Eq. (12), apply the local gap condition, and invoke Theorem 1. $\square$

Corollary 1 bounds worst-case spectral-efficiency loss under local compatibility between neural stationarity and the policy-space gap; the last term is the entropy-regularization error.

V. COMMUNICATION PERFORMANCE EVALUATION

We use worst-case normalized discounted spectral-efficiency return as the primary metric and unregularized exploitability Δ , field norm, and sampled-direction error as diagnostics.

A. Adversarial DSA Scenario and Evaluation Protocol

Wireless scenario: We evaluate the link and adaptive jammer in Fig. 2. The nominal benchmark uses $M = L = 8, 12$ fixed seeds, $\rho_h = 0.75$, $N_0 = 1$, $I_t^{\text{pri}}(a) = 0$, $g_t^J(b) = 1$, $(P_s, P_J, c_{\text{sw}}) = (10, 20, 0.10)$, $\delta = 0.95$, and $\tau = 0.02$. Its base secondary-link gain vector is $g_m^{\text{base}} = 1.9(0.18/1.9)^{m/(M-1)}$, $m = 0, \dots, M-1$, and the $L = M$ radio modes are cyclic shifts of this vector. The jammer-to-noise ratio is

$\rho_J = P_J/N_0$ and equals 20 nominally; adjacent-channel leakage $\lambda = 0.80$ enters the cyclic profile $1, \lambda, 0.12, 0.025$ at channel distances $0, 1, 2, >2$. The channel-scale study uses $M = L \in \{6, 8, 10, 12\}$, while the 12-seed wireless-stress study uses $M = L = 6$ and varies ρ_J , c_{sw} , λ , and ρ_h one factor at a time.

Policies and comparison protocol: Both players use independent two-layer tanh–softmax policies with the fully observed state encoding in Section II. The nominal setting uses 64 hidden units, 1608 parameters per player, and 30 joint updates; stress tests use 32 units and 25 updates. The nominal benchmark compares simultaneous policy gradient (SimPG), extragradient policy gradient (ExtraPG), a three-step PPM approximation (PPM-3), minimax proximal policy optimization (Minimax PPO), LCMPL-F, and LCMPL. They share the environment, policies, initialization, test seeds, entropy coefficient, outer-update count, and simultaneous start without warm-up; the five saddle-field rules also share the entropy-regularized policy-gradient oracle. Minimax PPO is a population clipped-surrogate reference with root-mean-square-normalized exact advantages and three clip-0.2 epochs of fresh per-update Adam [15]. Disjoint validation seeds select rates 0.01 for the fixed-step rules and 10^{-3} for Minimax PPO from prespecified grids. LCMPL-F is the field-only ablation defined in Section III-B.

LCMPL configuration: All LCMPL runs use the step-size box $[0, 0.08] \times [0, 0.12]$, merit weights $(\lambda_\Delta, \lambda_R) = (0.35, 1.5)$, and component normalization by positive initial values. The released configuration records the directional-model scales and candidate-acceptance tolerances used in every regime.

B. Nominal Anti-Jamming Performance

LCMPL reaches a final mean worst-case rate of 1.1106 bit/s/Hz in Fig. 4(a), versus 1.0380 for Minimax PPO. Against LCMPL-F, its paired rate gain is 0.0102 bit/s/Hz (95% confidence interval $[0.0006, 0.0197]$), and its exploitability reduction is 0.0728 ($[0.0346, 0.1109]$). Minimax PPO has smaller optimization residuals but a lower rate; stationarity alone does not ensure robust communication.

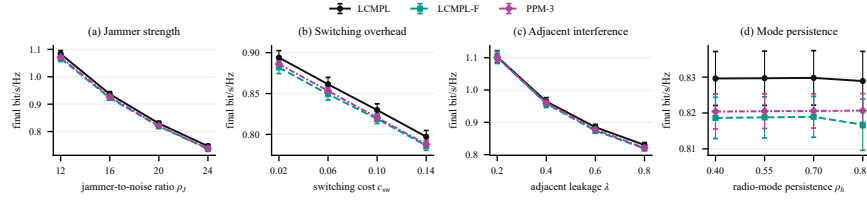


Fig. 5. Wireless stress across jammer strength, switching cost, leakage, and mode persistence: 12-seed means and 95% confidence intervals.

C. Channel Scalability and Sampled-Direction Fidelity

Fig. 4(f) shows that LCMPL exceeds LCMPL-F in final mean worst-case rate across 6–12 channels. Panel (e) reports sampled field and Jacobian–vector-product (JVP) errors against population values at three checkpoints. Increasing the batch from 32 to 256 trajectories reduces mean relative field and JVP errors by 64.7% and 86.9%, respectively. The decline supports the conditional learning-residual model used in Theorem 1.

D. Trajectory-Sampled Anti-Jamming Learning

Panels (g) and (h) evaluate a trajectory-sampled implementation of LCMPL under an equal interaction budget. Each update uses 128 length-64 on-policy trajectories, and every method receives 245,760 transitions per seed. Both players share one batch; automatic differentiation forms the same-batch Jacobian–field direction, while ExtraPG and PPM-3 reuse the batch through per-decision likelihood ratios. Exact finite-game dynamic programming evaluates the merit, same-state acceptance test, and evaluation checkpoints. At the final budget, LCMPL improves worst-case rate over LCMPL-F by 0.0223 bit/s/Hz with a paired 95% confidence interval of $[0.0138, 0.0308]$ and reduces exploitability by 0.0672 with interval $[0.0440, 0.0903]$. LCMPL also attains a higher rate and lower exploitability than SimPG, ExtraPG, and PPM-3. The accepted update has $\gamma_k > 10^{-10}$ on 25.3% of 360 updates, while the field-only candidate is selected on another 24.2%. The rate gain with selective curvature activation agrees with the local-geometry mechanism rather than an always-on second-order correction.

E. Robustness Across Wireless Operating Points

Fig. 4(d) shows 12-seed leakage-dependent curvature activation, while Fig. 5 retrains LCMPL, LCMPL-F, and PPM-3 under one-factor changes in jammer-to-noise ratio, switching cost, leakage, and mode persistence. LCMPL has the highest final mean rate in every plotted sweep cell, with paired mean gains of 0.0019–0.0161 bit/s/Hz over LCMPL-F. The advantage across all four stress axes shows that the step-size controller is not tied to the nominal jammer or fading regime.

VI. CONCLUSION

LCMPL separates field and curvature step sizes in adversarial DSA and retains curvature only when the same-state acceptance test validates its merit and rate. The learning-error analysis bounds time-averaged stationarity and, under a local policy-gap condition, worst-case spectral-efficiency loss; LCMPL achieves the highest final mean rate across the reported regimes.

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